

Origins and Evolution of Language

Week 2: Natural selection, adaptation, and the evolution of language

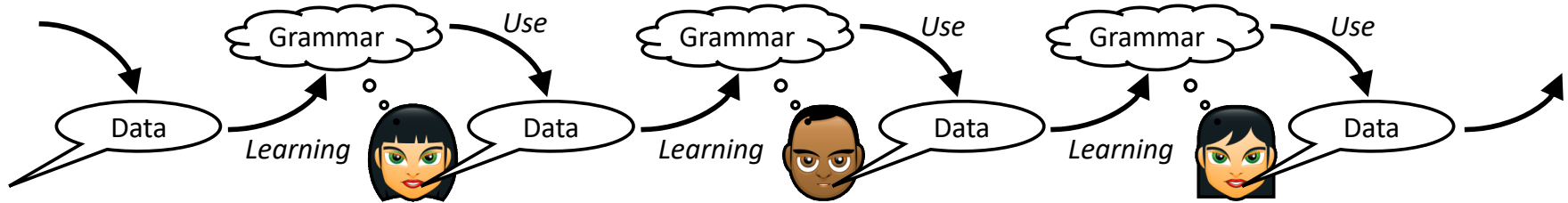
Kenny Smith

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Workshops start this week

- Materials have been online since last Tuesday
- Attendance recorded and compulsory
- Know where you are going
 - Wednesday morning room is hard to find!
- Take a look at the exercise in advance (less intensive prep this week than usual)
- **Bring a laptop!**
- Tutor notes released after the Thursday workshop

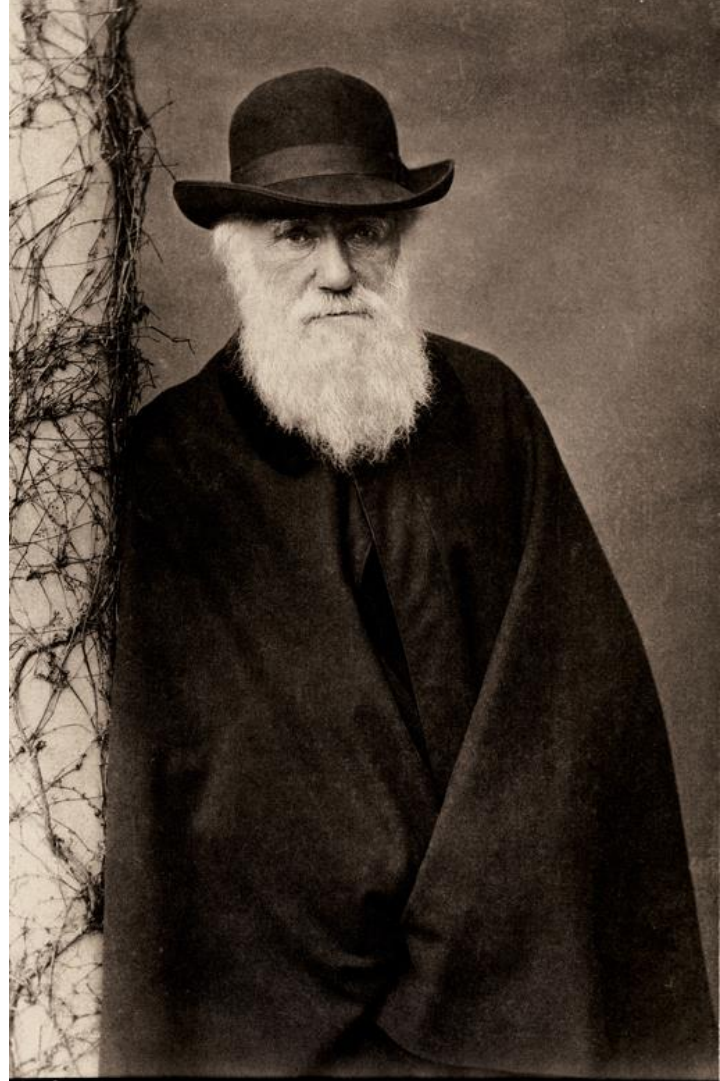
Learning, use, and language design



- Language is passed from person to person by **learning**
- People learn from language as it is **used in communication**
- Language **evolves** in response to its learning and use
- Linguistic structure is a product of this evolutionary process

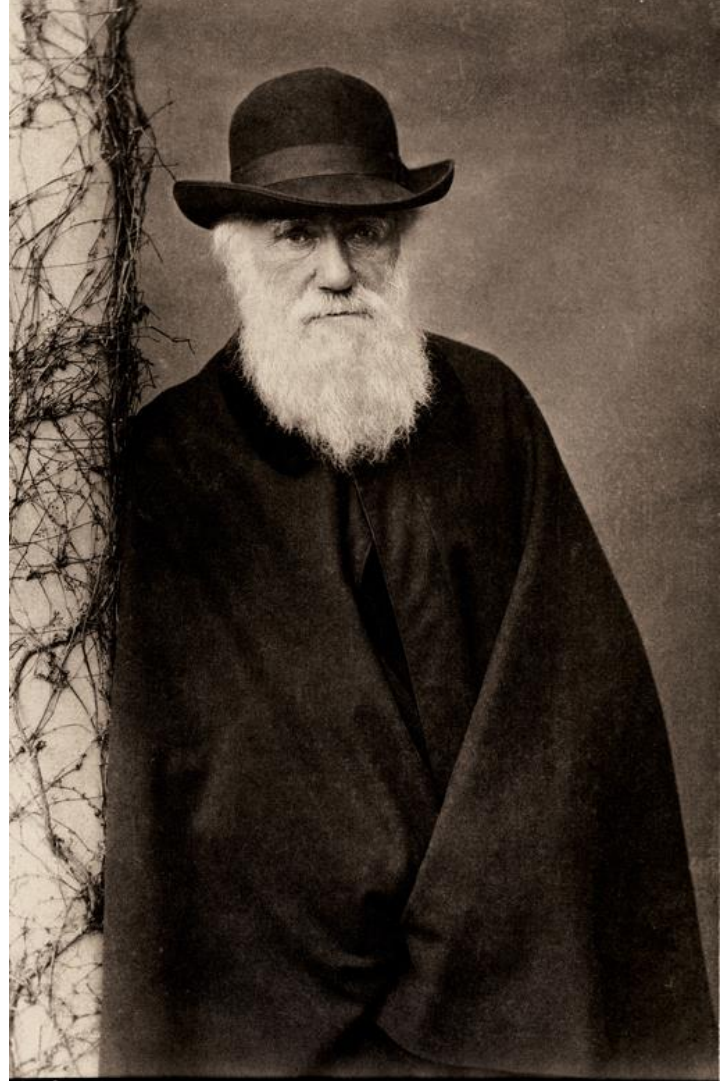
From last week

- Language is just like any other adaptive feature of an organism's biology
- It's an **innate** feature of the human mind
- It evolved by natural selection under pressure for communication



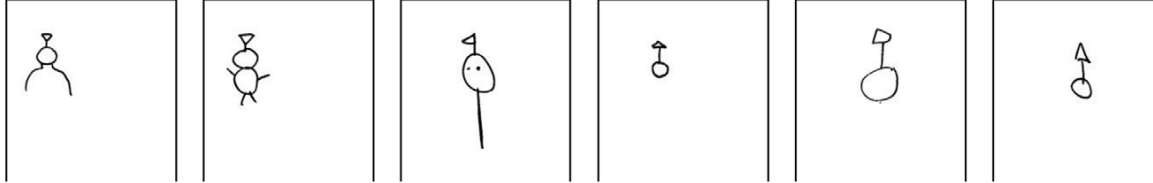
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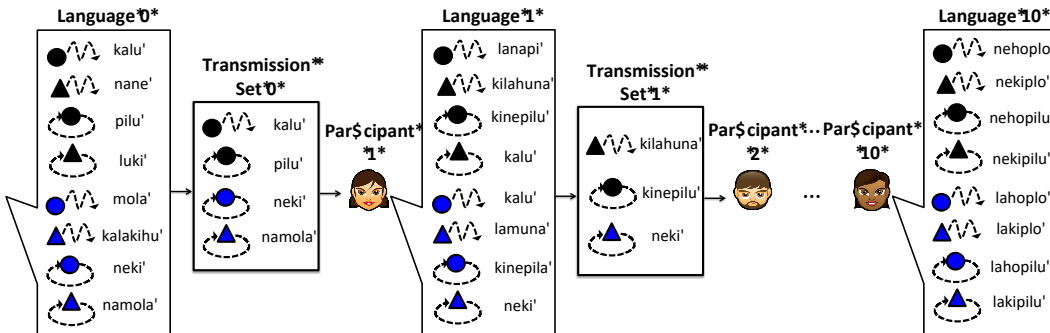
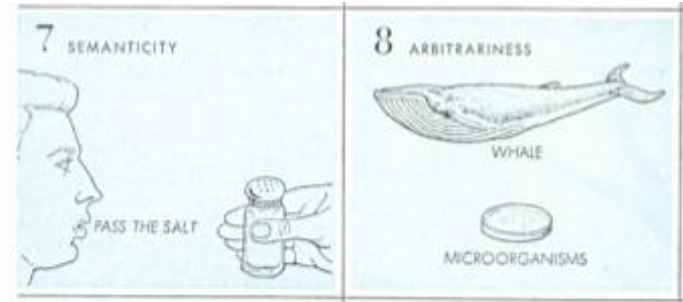


Wrap up from last week (leading in to your questions)

Participant 1 (Round 1) **Participant 2 (Round 2)** **Participant 3 (Round 3)** **Participant 4 (Round 4)** **Participant 5 (Round 5)** **Participant 6 (Round 6)**



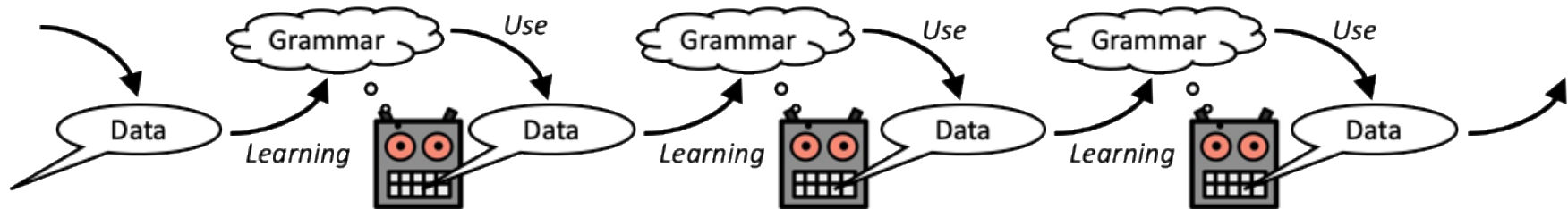
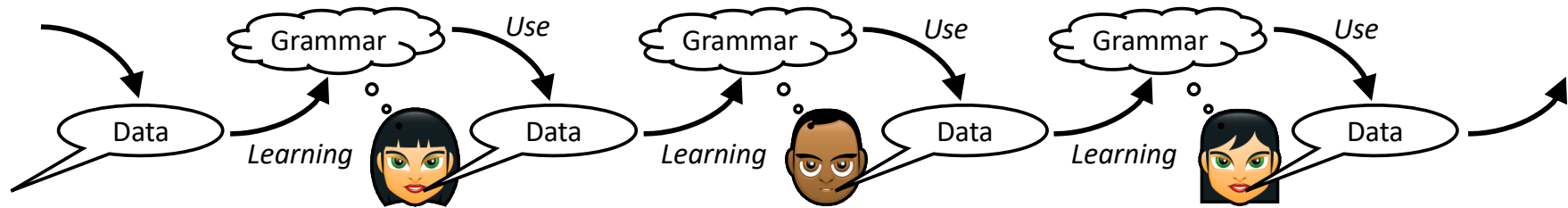
Caldwell, C. A., & Smith, K. (2012). Cultural evolution and the perpetuation of arbitrary communicative conventions in experimental microsocieties. *PLoS ONE*, 7, e43807.



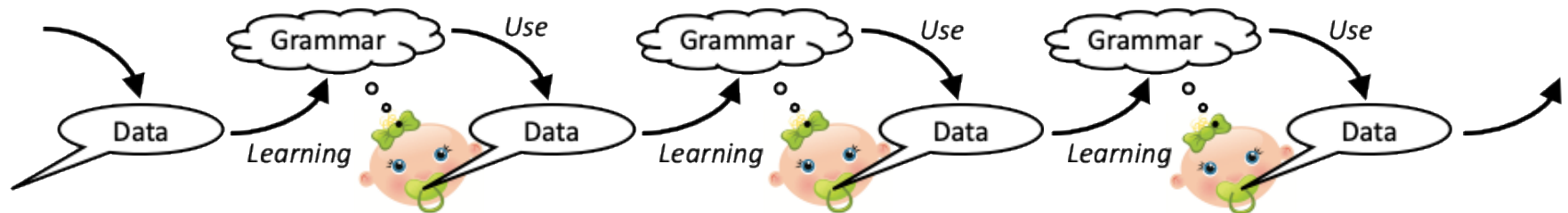
Kirby, S., Cornish, H., & Smith, K. (2008). Cumulative cultural evolution in the laboratory: an experimental approach to the origins of structure in human language. *Proceedings of the National Academy of Sciences, USA*, 105, 10681-10686.



“To what extent do these results depend on the fact that these people already have a language?”



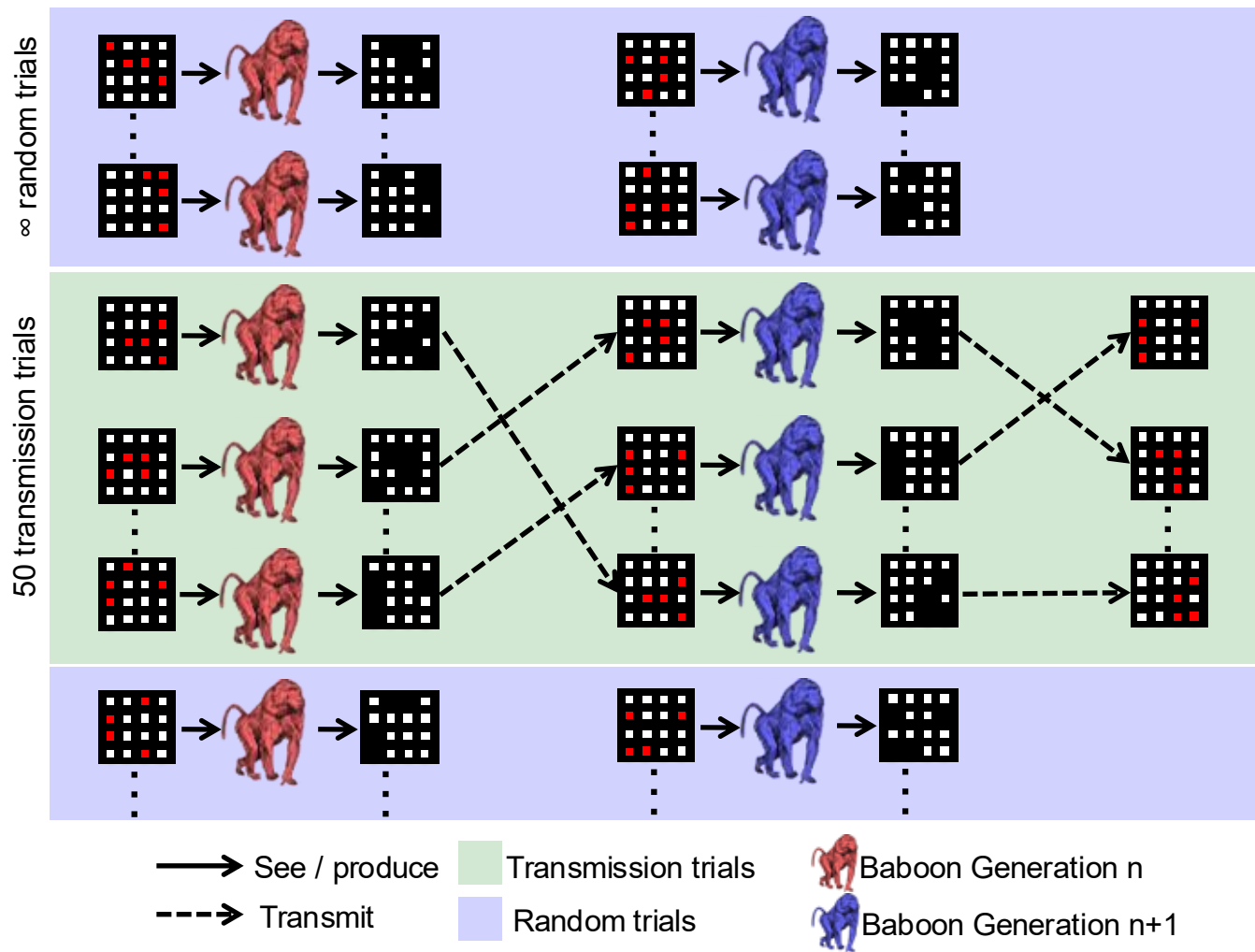
Methods taught in Simulating Language (Semester 2)

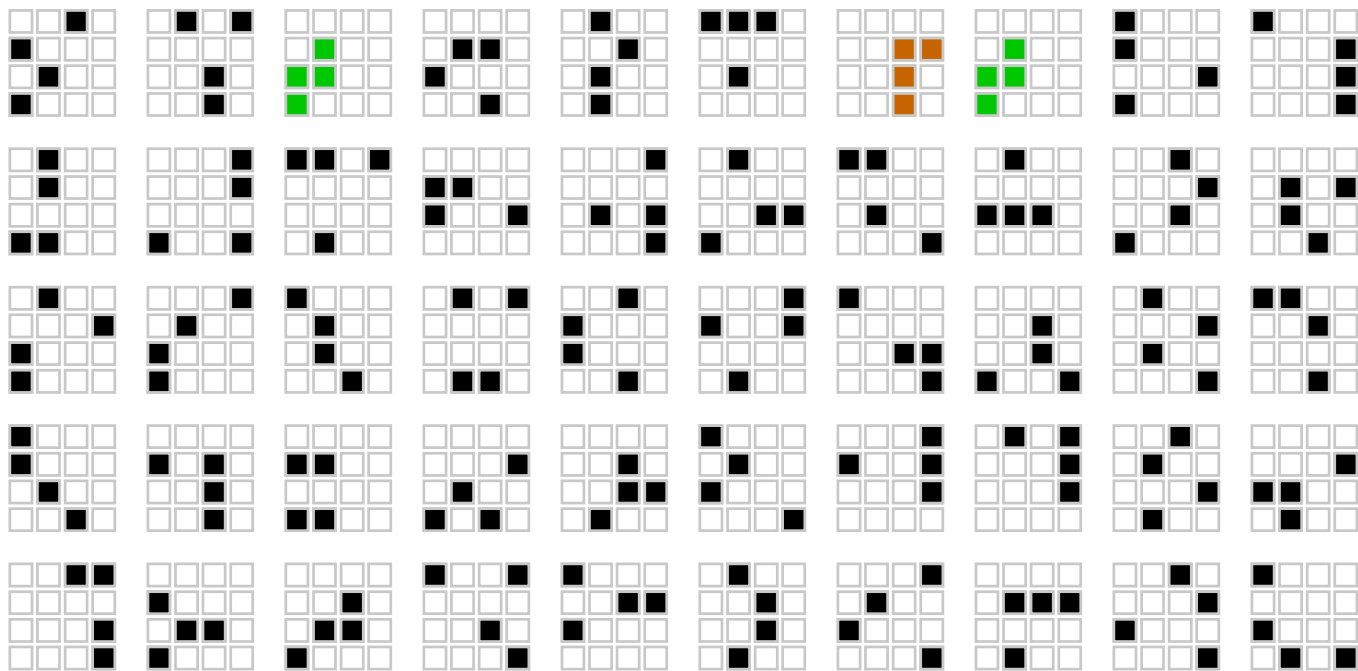


Raviv, L., & Arnon, I. (2018). Systematicity, but not compositionality: Examining the emergence of linguistic structure in children and adults using iterated learning. *Cognition*, 181, 160-173.

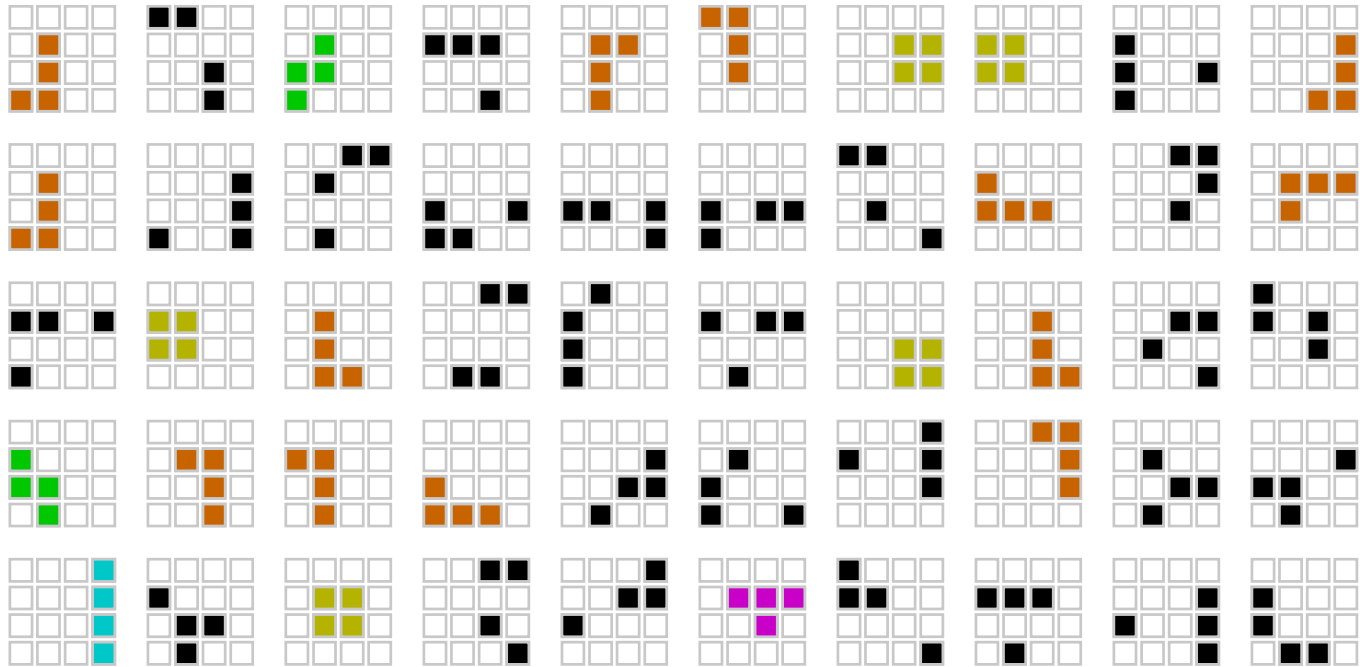




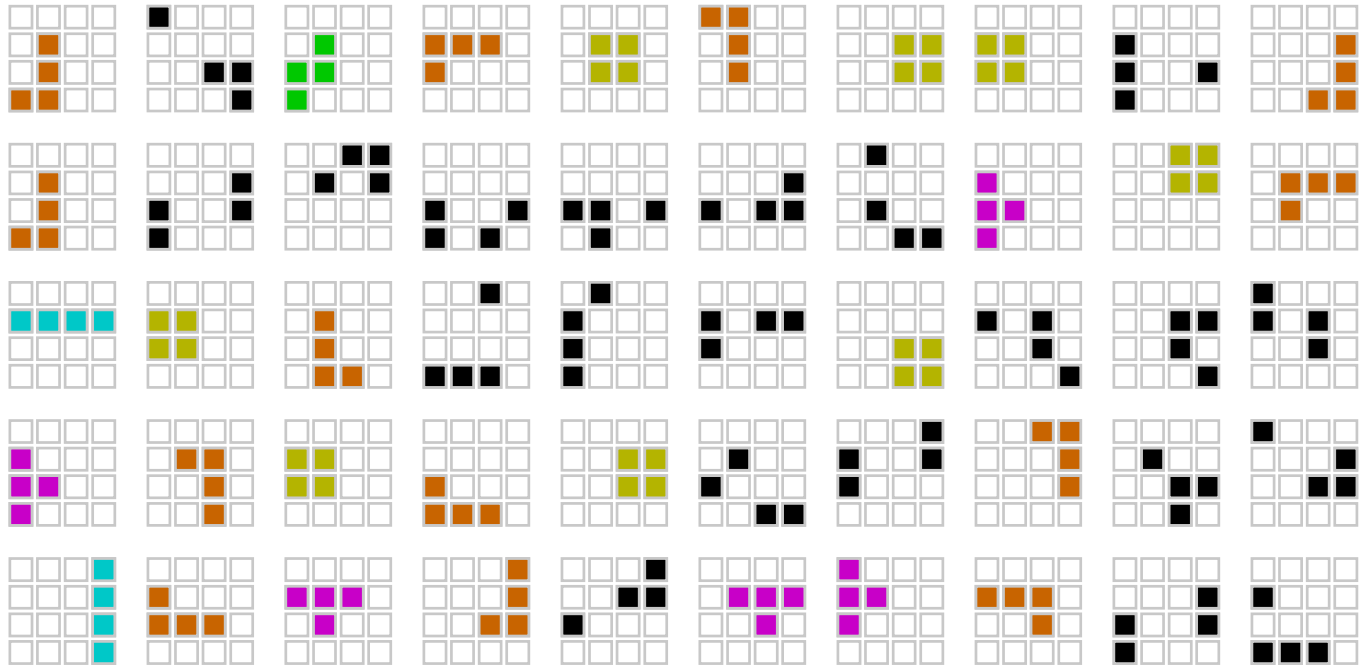




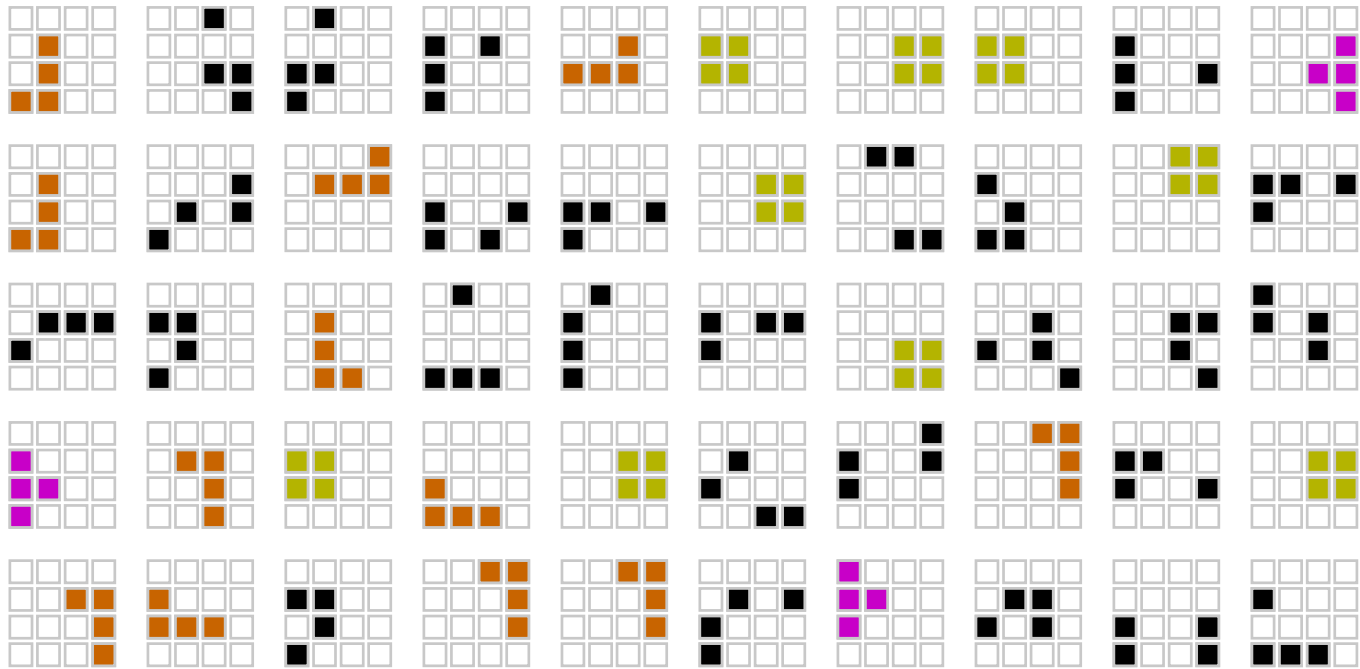
Random grids



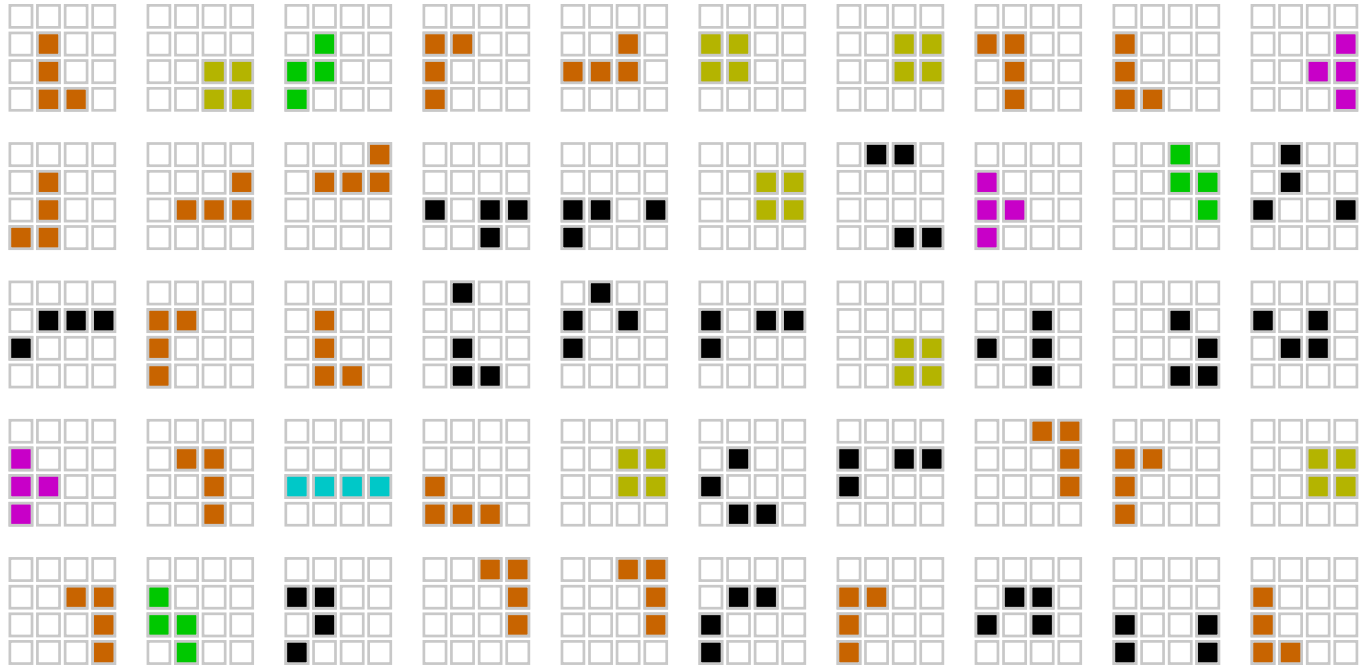
Generation 1



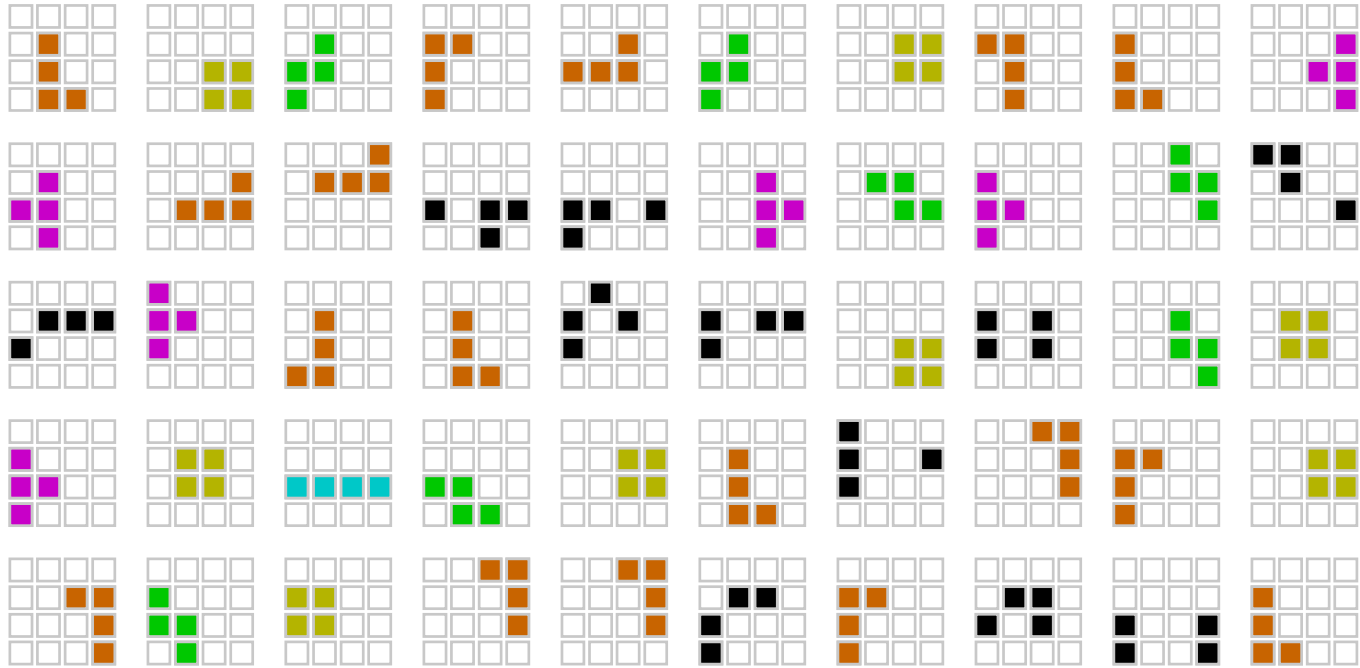
Generation 2



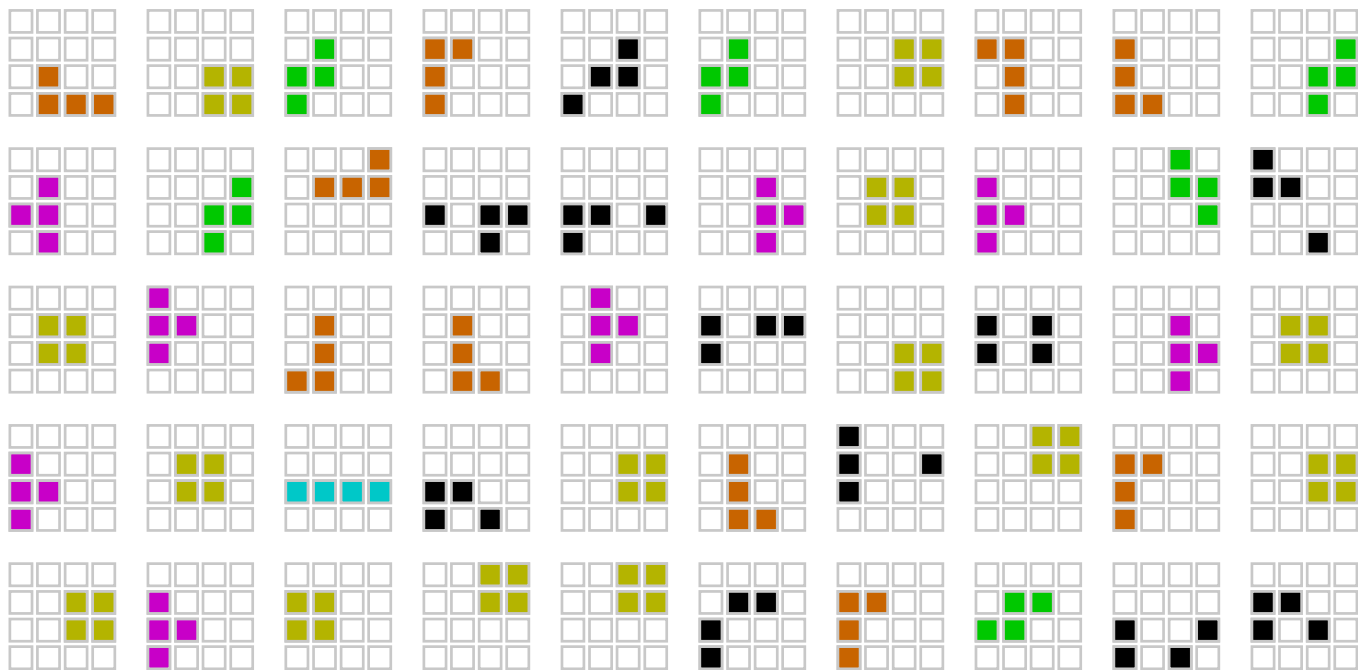
Generation 3



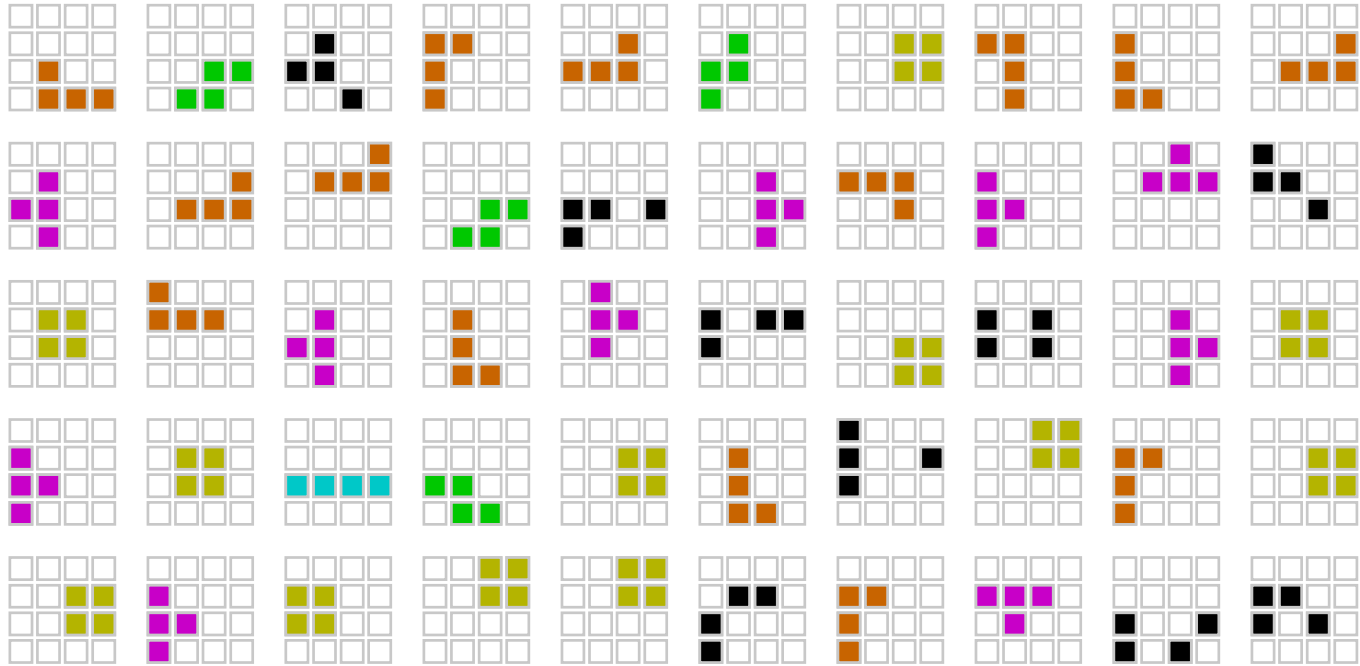
Generation 4



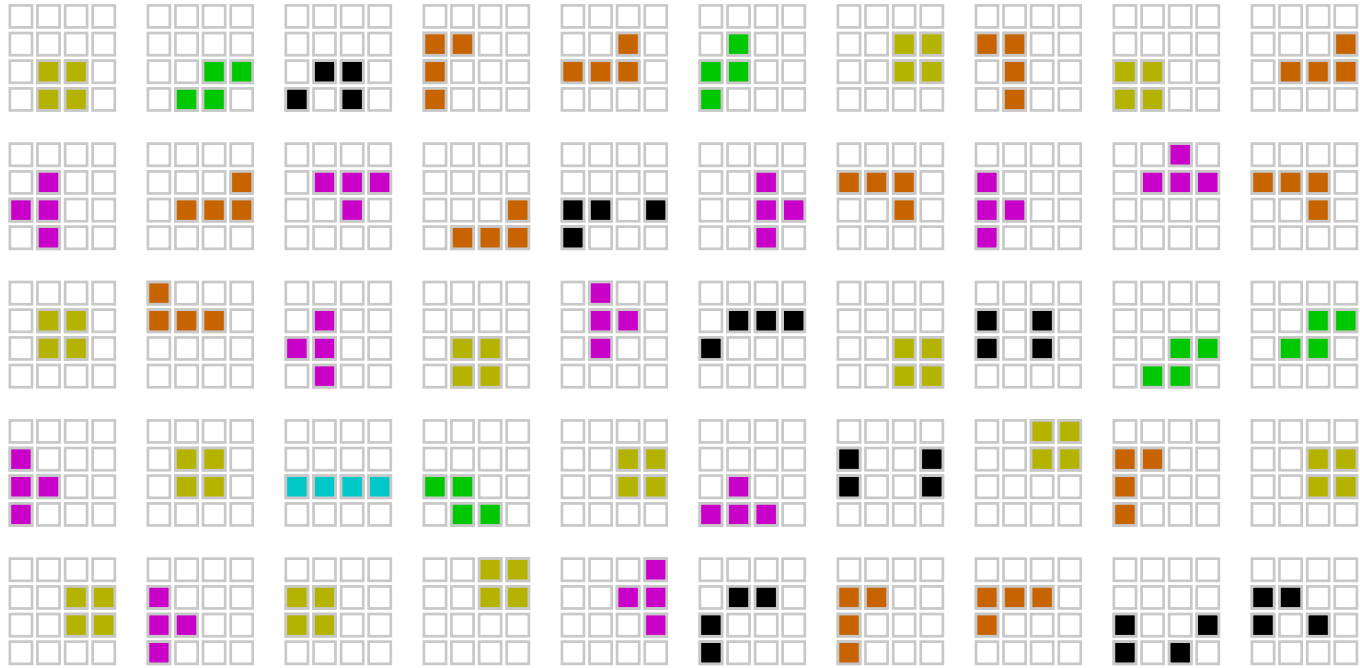
Generation 5



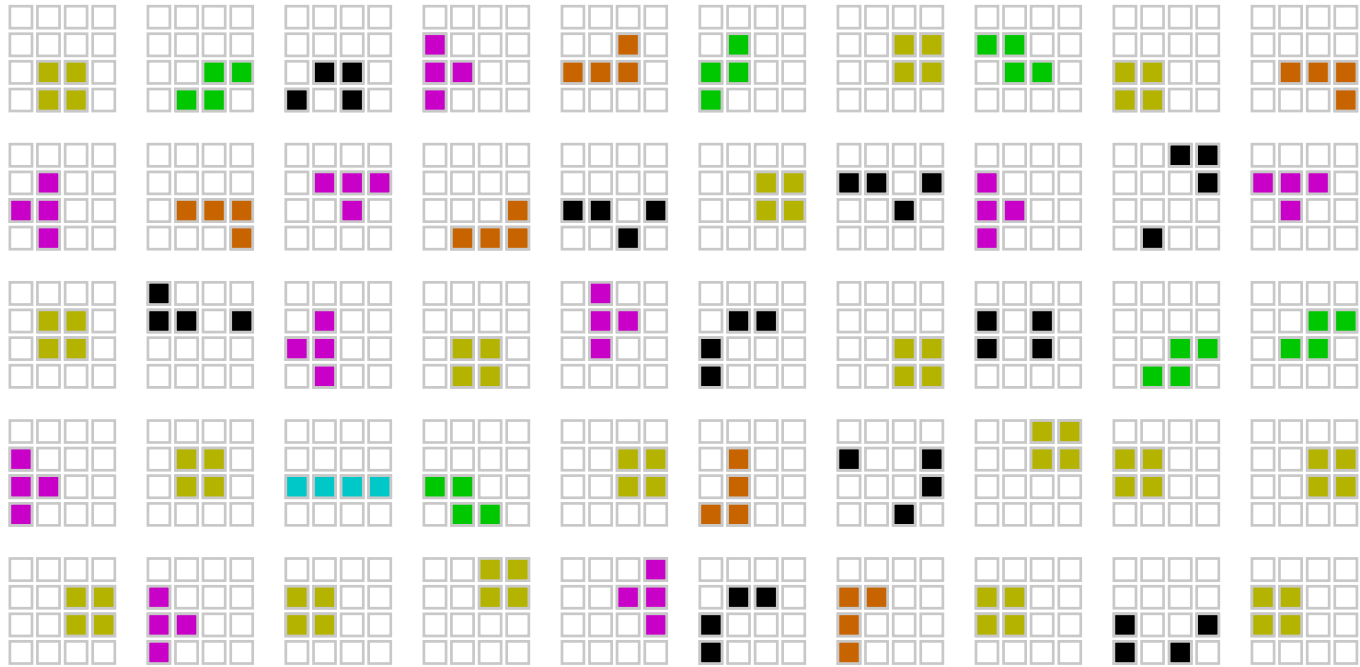
Generation 6



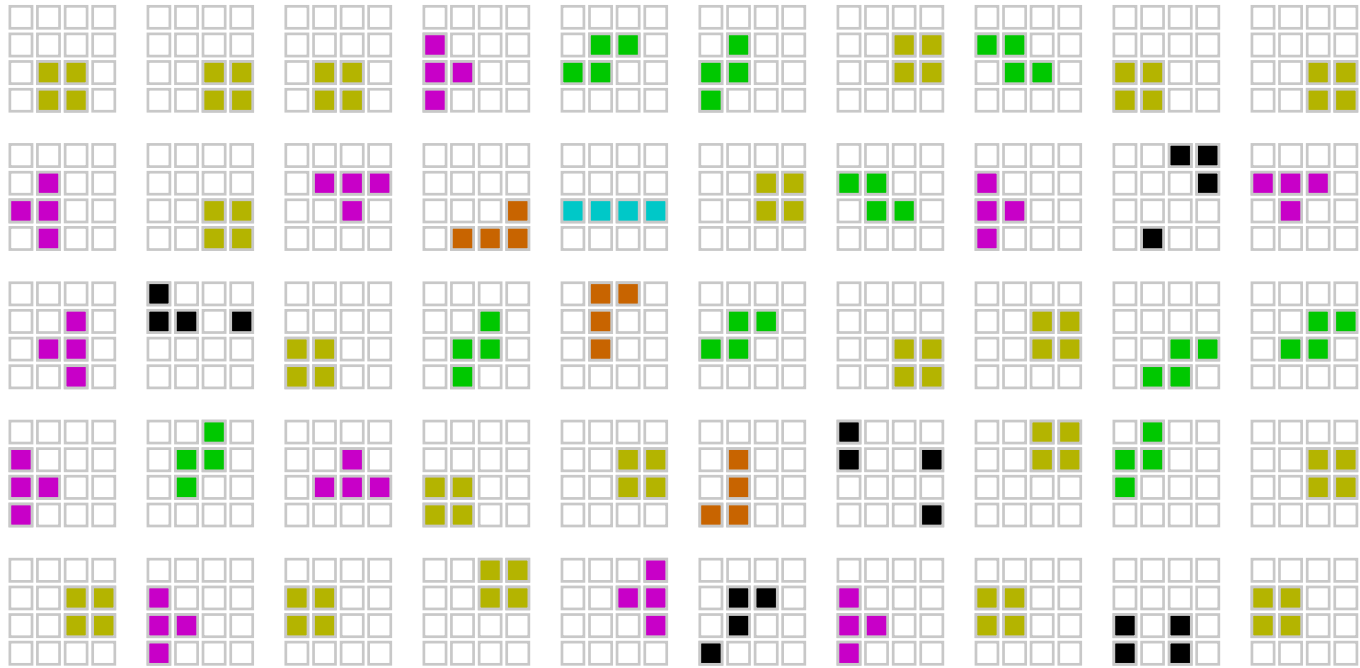
Generation 7



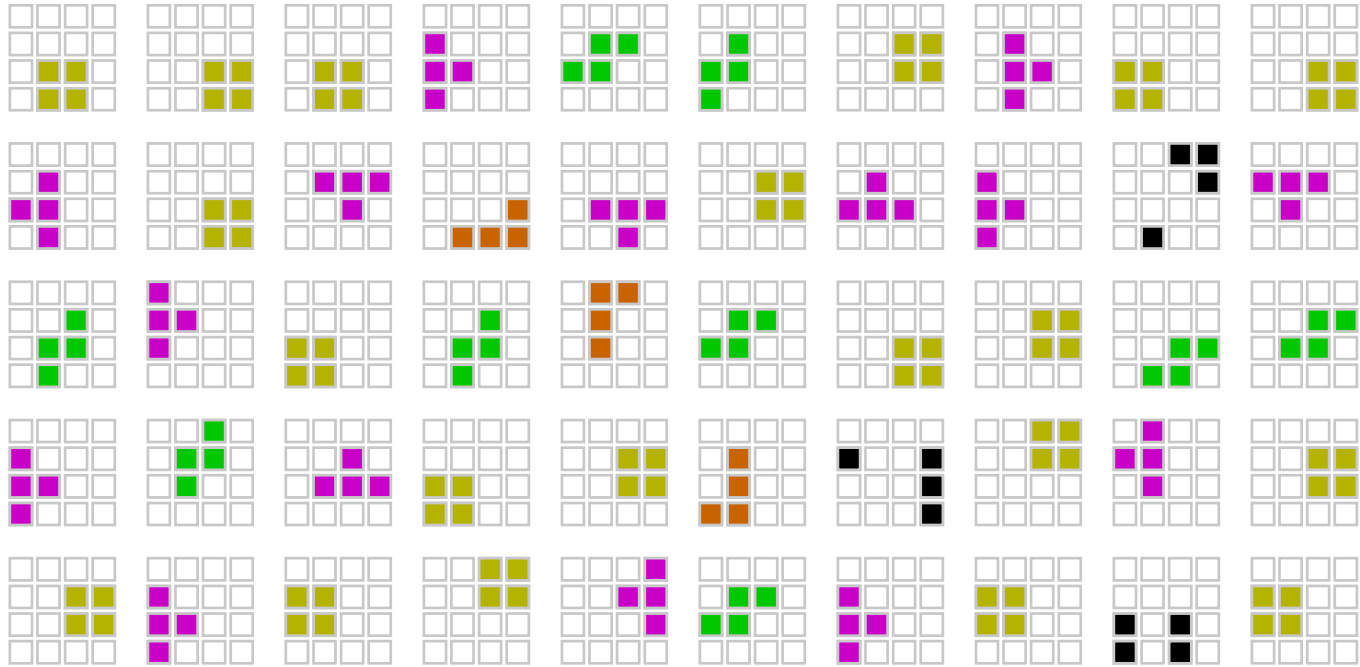
Generation 8



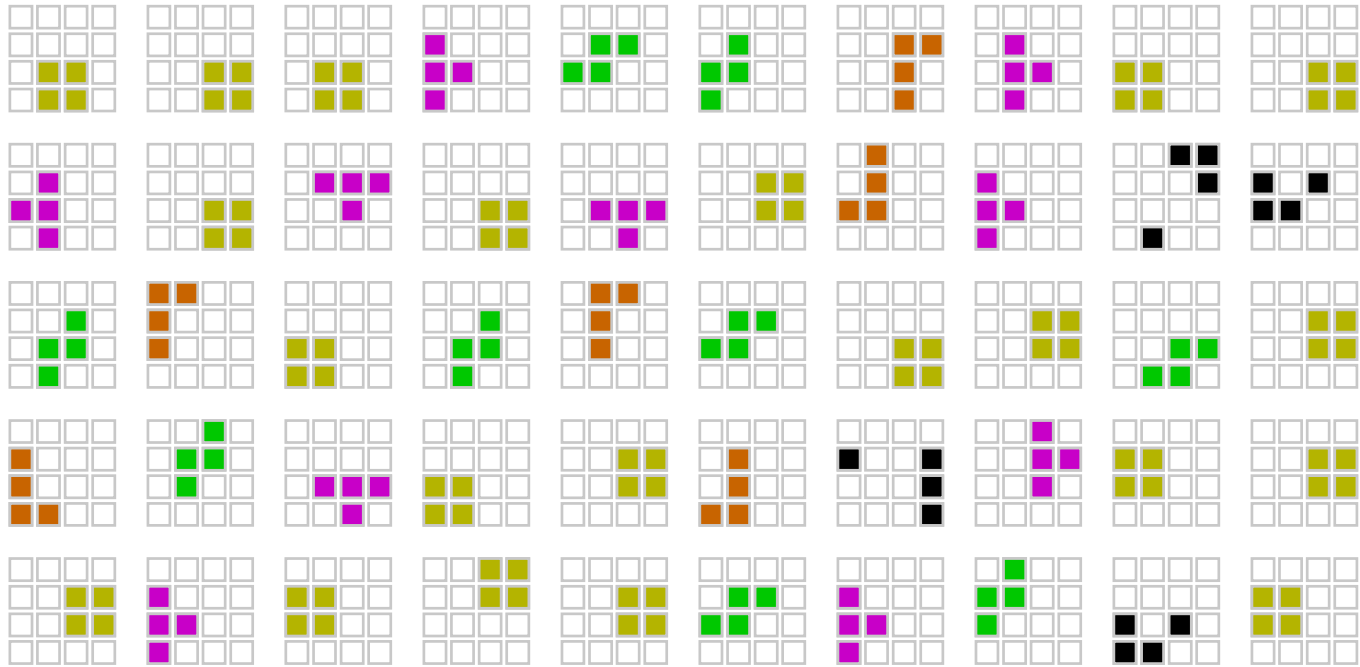
Generation 9



Generation 10



Generation 11



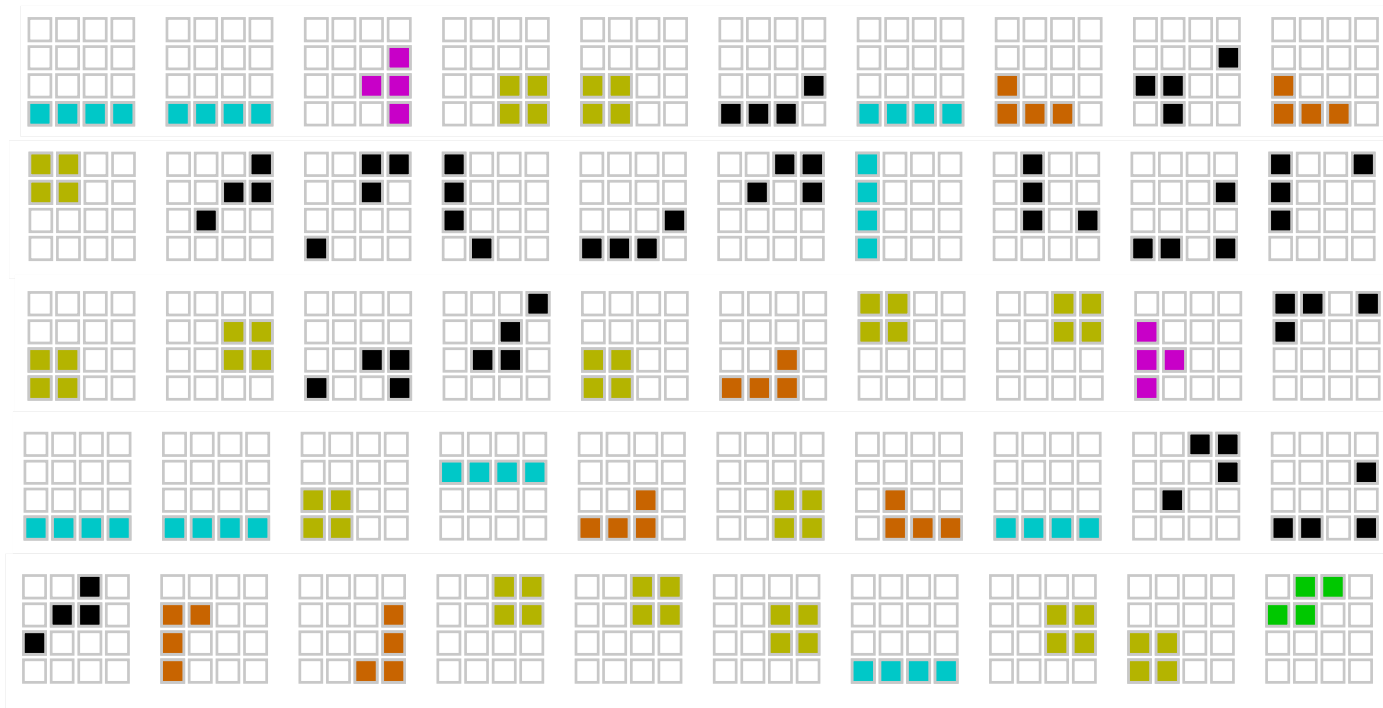
Generation 12

Emergence of a **system**



Chain 4, Generation 12

Emergence of a **system**



Chain 1, Generation 12

Systematic structure develops even
in baboons (if you scaffold their
environment in the right way)

Your questions

“when I was trying to watch back the lecture recording from last week I couldn’t hear any sound. ”

“While trying to imitate language acquisition [and language evolution through iterated learning!] in the lab is useful in certain ways, this is always going to be second-language acquisition. How do these studies deal with the fact that they're not really able to recreate first-language acquisition?”

“Since the species most closely related to us are now extinct, are there any theories about whether these species were/would be capable of language like ours? Obviously, it cannot be traced or proved; just wondering if anyone is researching this or if there are any theories.”

- We'll talk about this in week 5
- For a controversial (?) take: Dediu, D. & Levinson, S. C. (2013) On the antiquity of language: the reinterpretation of Neandertal linguistic capacities and its consequences. *Frontiers in Psychology*, 4, 397.

<https://doi.org/10.3389/fpsyg.2013.00397>

“I am curious about how do you response to the theological explanations of the origin of languages. Yet, many scholars might think that it is out-dated and non-scientific; nevertheless, I feel there are many thoughts we may borrow to inspire the relevant researches.”

“How are information and language related? Is there information outside of language?

Can language be understood as the ‘explicit making’ of ideas?

Are there limits to what language can express?”

“What is the difference between convergent evolution and analogies?”

- Covered today

“What exactly is the cognitive niche theory?”

“Tooby and DeVore proposed that the human lifestyle is a consequence of a specialization for overcoming the evolutionary fixed defences of plants and animals ... by cause-and-effect reasoning. Such reasoning enables humans to invent and use new technologies ... that exploit other living things before they can develop defensive countermeasures in evolutionary time.”

“three key features of the distinctively human life-style – knowhow, sociality, and language—co-evolved, each constituting a selection pressure for the others” (Pinker, 2003, p.29)

“Going over question 8 in the lecture could help. I am not sure I understood it correctly”

Question 8: [This is an exam-style question] If language is an adaptation, a product of evolution by natural selection, it must meet the requirements for evolution by natural selection - it must exhibit **heritable variation in fitness**. Drawing on the arguments in Pinker (2003), or on your own evidence, to what extent do you think language exhibits heritable variation in fitness? 1 paragraph max please!

“To what depth should we know about the biological aspects e.g. which genes are needed? I am also wondering if we need to know about the alternative theories such as the Whorfian hypothesis?”

“I was wondering how to interpret the formula for fitness”

$$\bar{w} = \frac{\sum (n_i \times w_i)}{N}$$

You don't have to!

But:

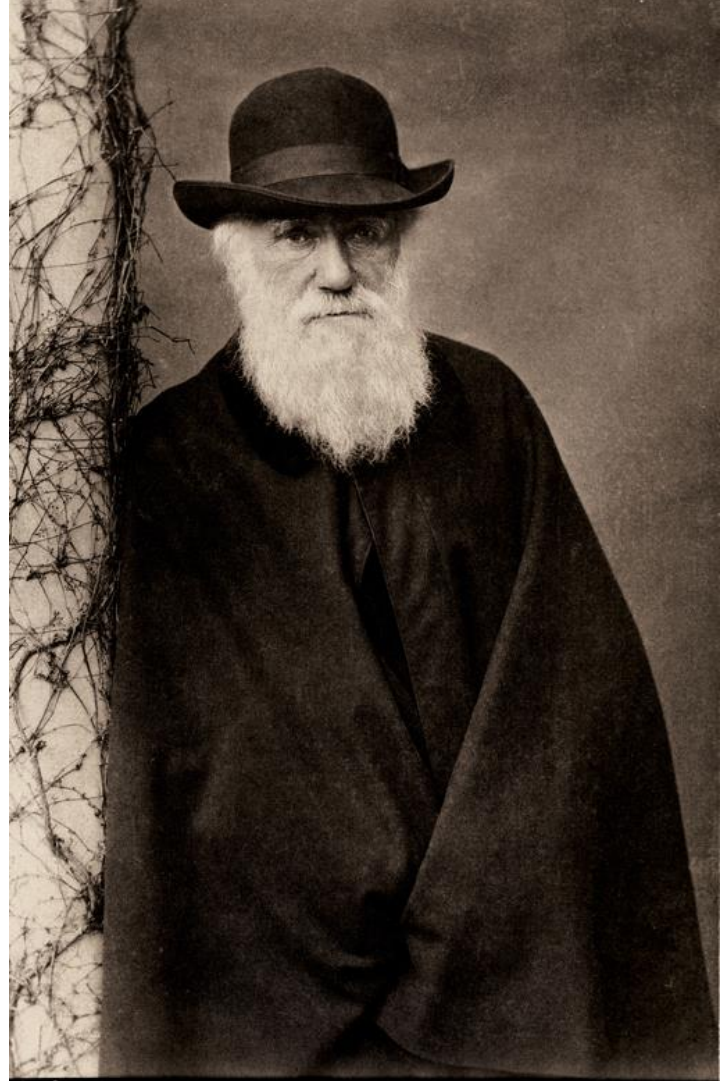
- n_i : number of individuals of type i
- w_i : number of **offspring** of individuals of type i (fitness)
- N : total population size
- $\bar{w}$: average number of offspring (average fitness)

The key thing: if type i is of above-average fitness, it will increase in proportion

Any more questions?

From last week

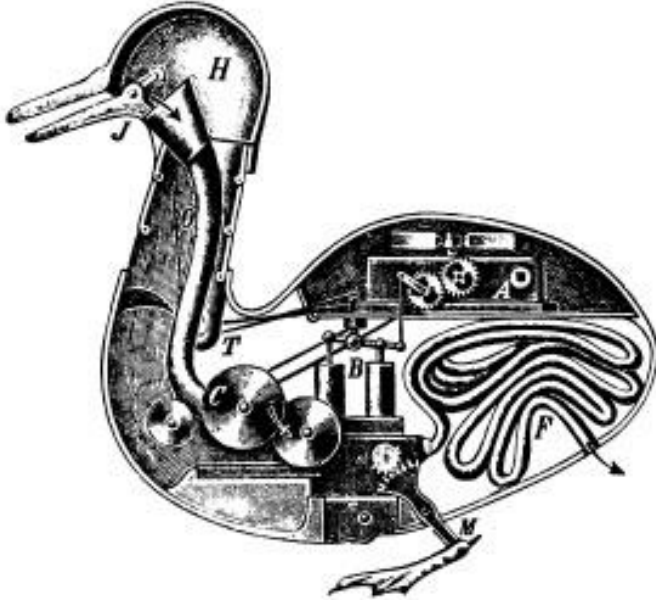
- Language is just like any other **adaptive** feature of an organism's biology
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Plan for the rest of today

- **Natural selection and adaptation**
 - Heritable variation in fitness
- Homology and analogy
- Language as an adaptation

Design in life



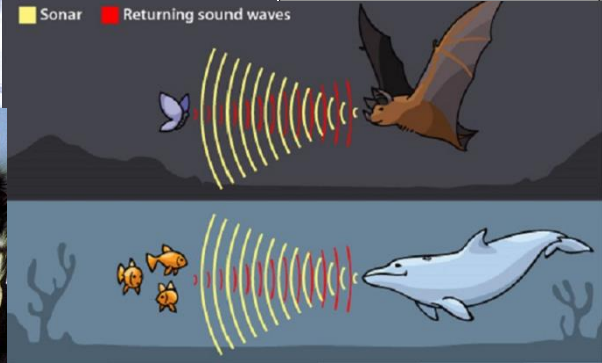
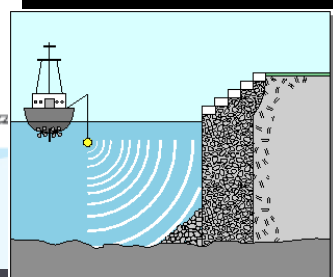
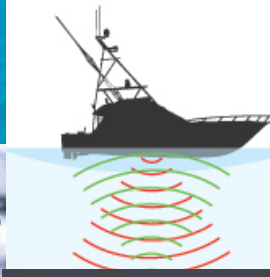
https://en.wikipedia.org/wiki/Digesting_Duck

Living things are remarkably competent

- They are self-sustaining, reproducing, autonomous, deal with hostile, changing environments, ...
- Designing a machine to do any of these things is currently beyond us



Design in life



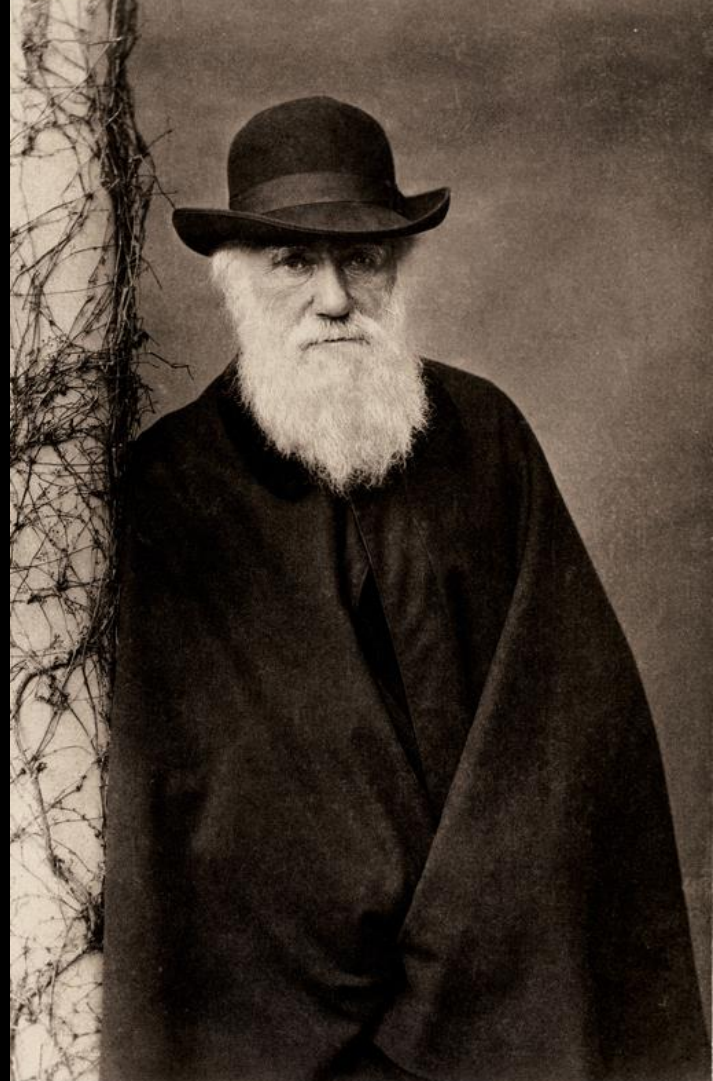
Adaptations

- “‘design’ in life – those properties of living things that enable them to survive and reproduce in nature.” (Ridley, 1996, *Evolution*, p. 5)
- “An adaptation is a characteristic that enhances the survival or reproduction of organisms that bear it” (Futuyama, 2009, *Evolution*, p. 279)
- “The process whereby the members of a population become better suited to some feature of their environment through change in a characteristic that affects their survival and reproduction” (Futuyama again, same page)
- “a characteristic that has evolved by natural selection” (Futuyama again, same page)

Evolution by natural selection

“Owing to this struggle for life, any variation, however slight and from whatever cause proceeding, if it be in any degree profitable to an individual of any species, in its infinitely complex relations to other organic beings and to external nature, will tend to the preservation of that individual, and will generally be inherited by its offspring. The offspring, also, will thus have a better chance of surviving, for, of the many individuals of any species which are periodically born, but a small number can survive. I have called this principle, by which each slight variation, if useful, is preserved, by the term of Natural Selection”

(Darwin, 1859/1965, *On the Origin of Species*, p. 61)



Evolution by natural selection

“Natural selection is easiest to understand, in the abstract, as a logical *argument*, leading from premises to conclusion...

1. Reproduction. Entities must reproduce to form a new generation.
2. Heredity. Offspring must tend to resemble their parents: roughly speaking, “like must produce like.”
3. Variation in individual characters among members of a population. ...
4. Variation in the *fitness* of *organisms* according to the state they have for a heritable character. ... individuals in the population with some characters must be more likely to reproduce (i.e. have higher fitness) than others

If these conditions are met, for any property or species, natural selection *automatically* results. If any conditions are not met, natural selection does not result. ... When all four conditions apply, the entities with the property conferring higher fitness will leave more offspring, and the frequency of that type of entity will increase in the population” . (Ridley, *Evolution*, p. 71-72)

A nasty quiz question many of you got wrong

Go to wooclap.com, use code INMRBS

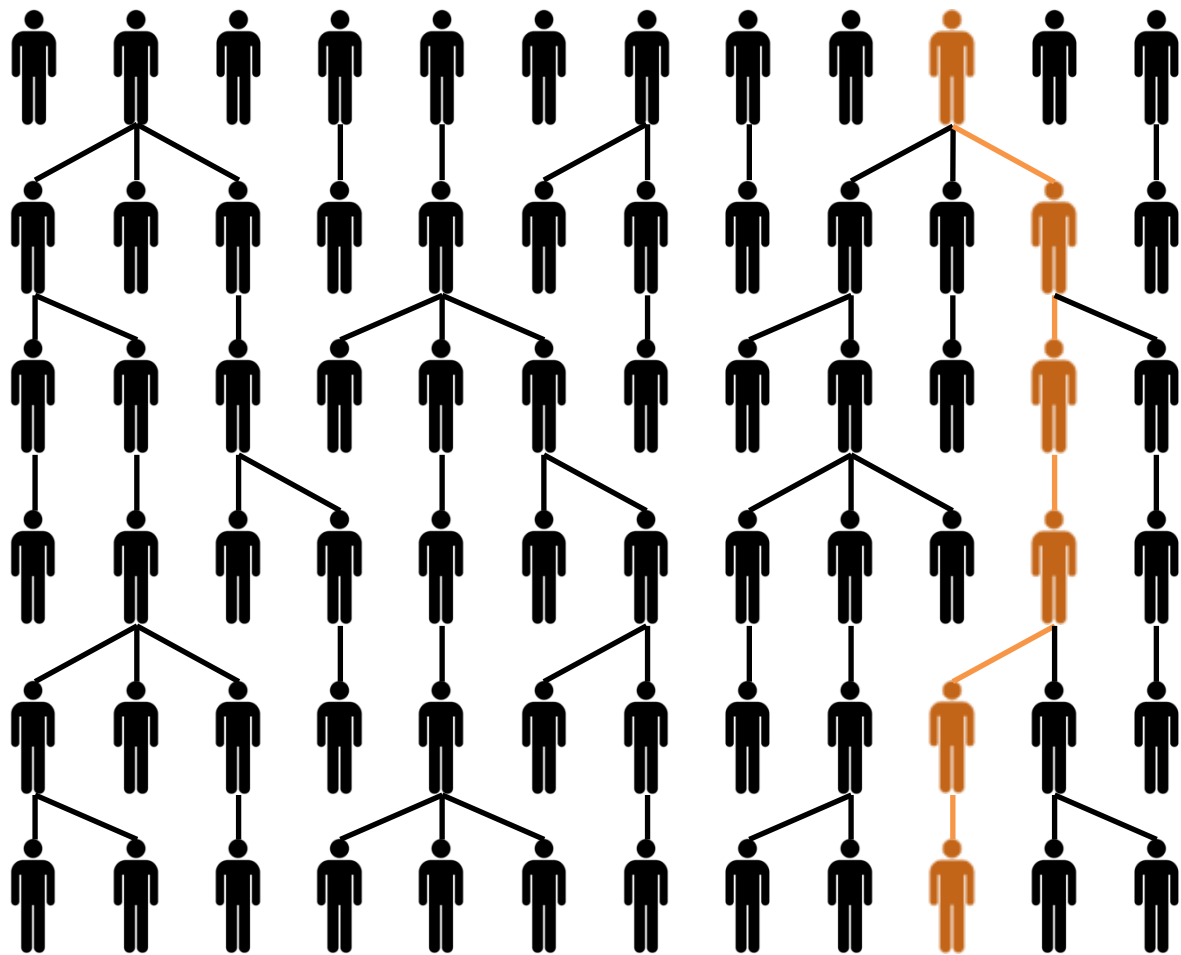
Question 2: Which if the following is **not** required for natural selection to take place?

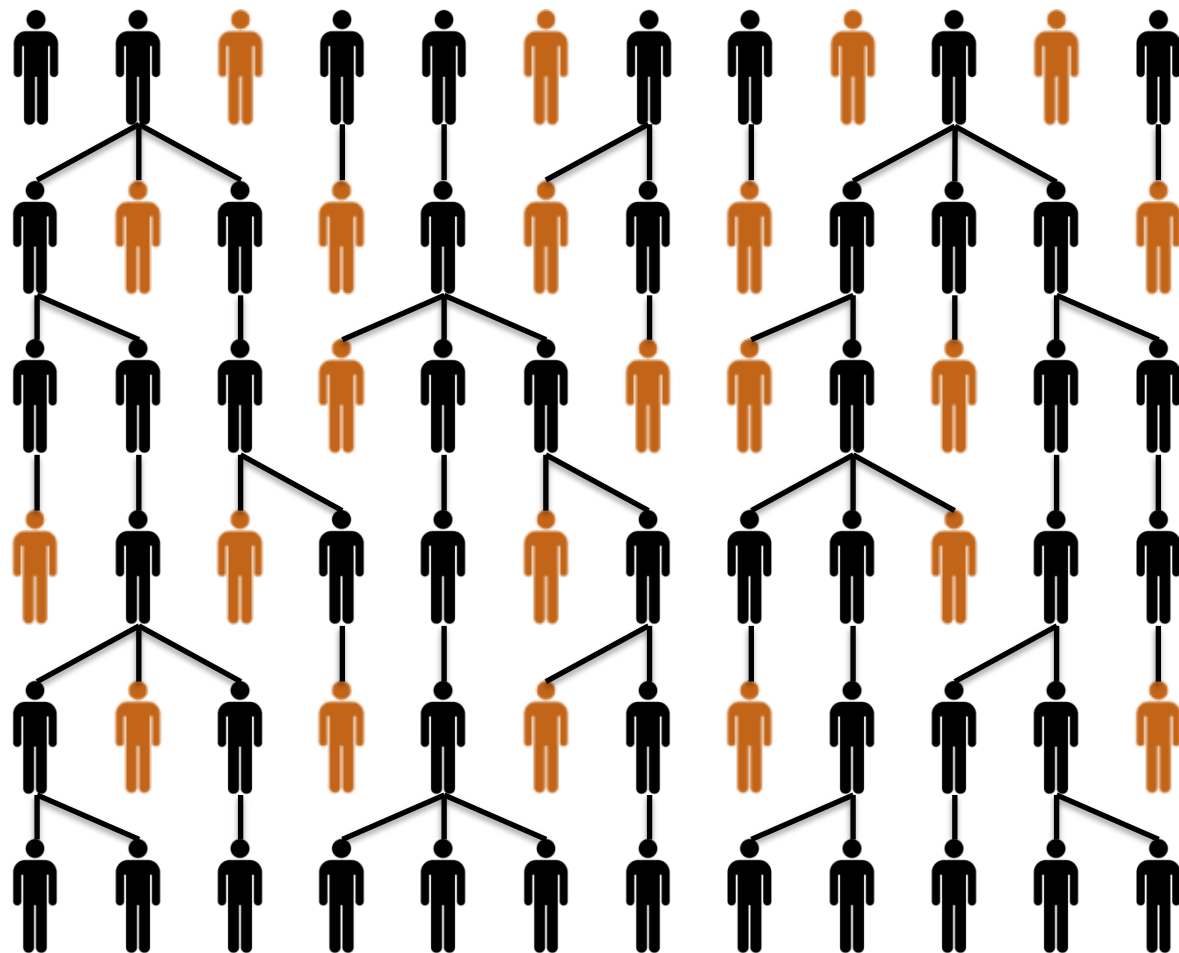
1. Organisms must differ in their ability to survive
2. Organisms must differ in the number of offspring they produce
3. Variation
4. Offspring must resemble their parents

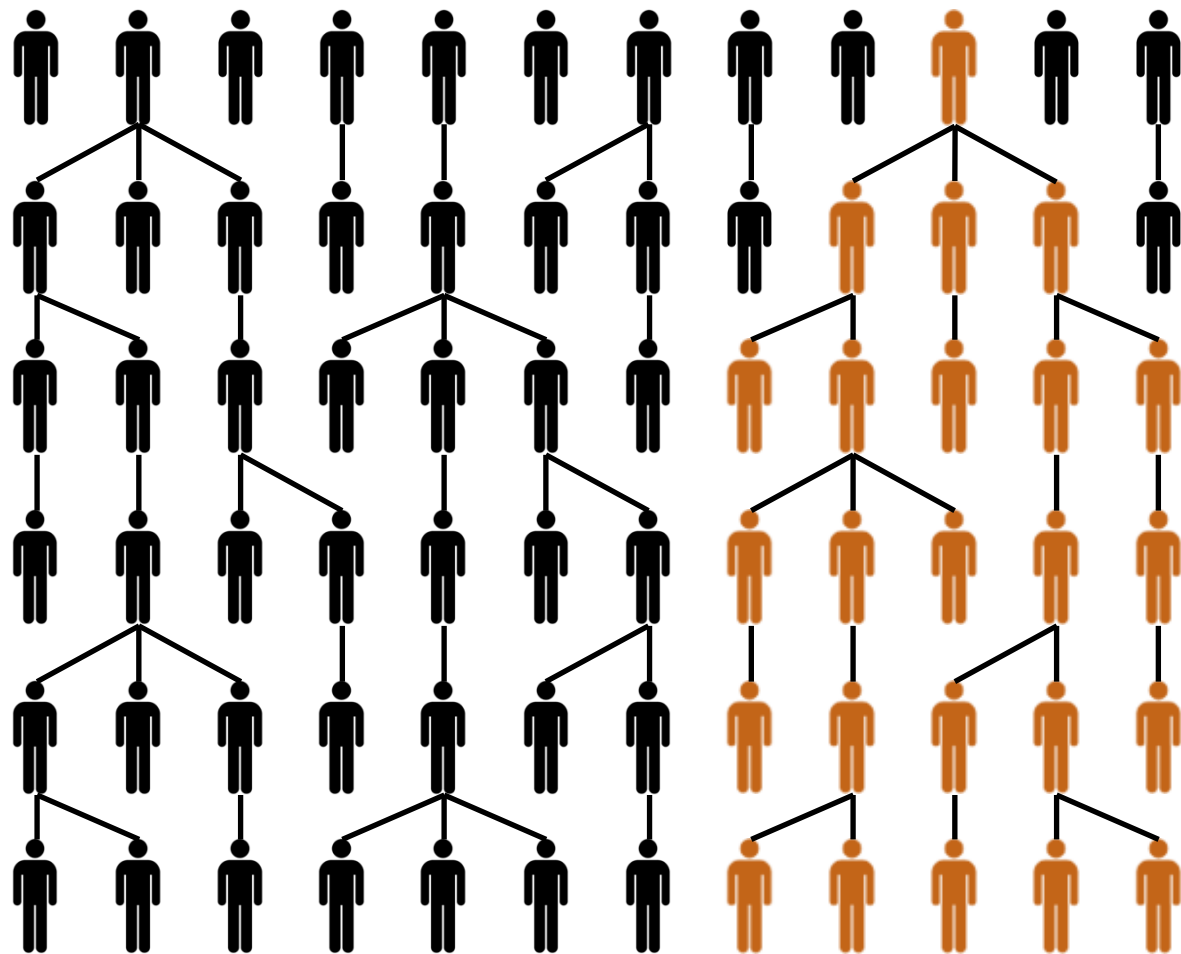
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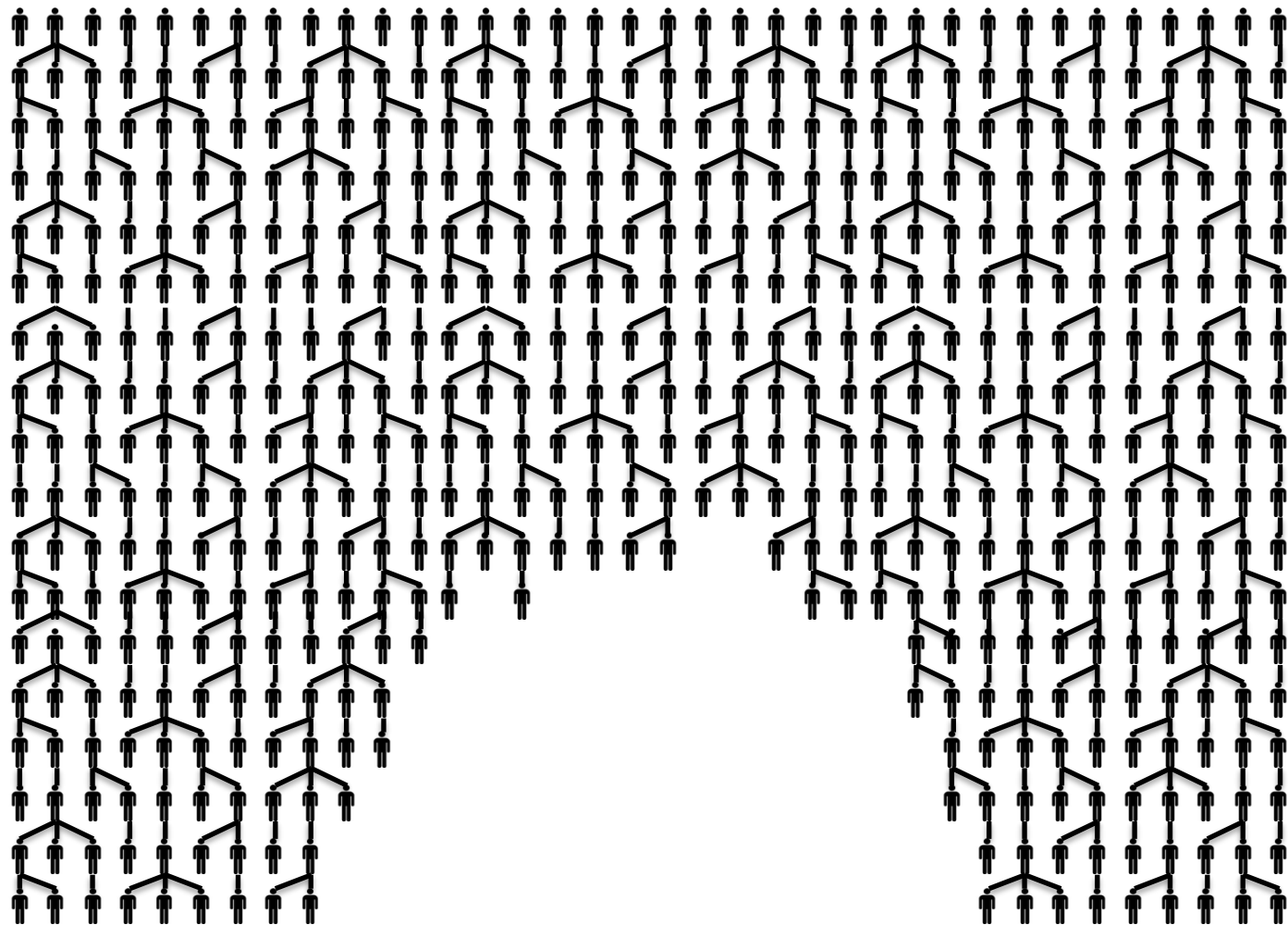
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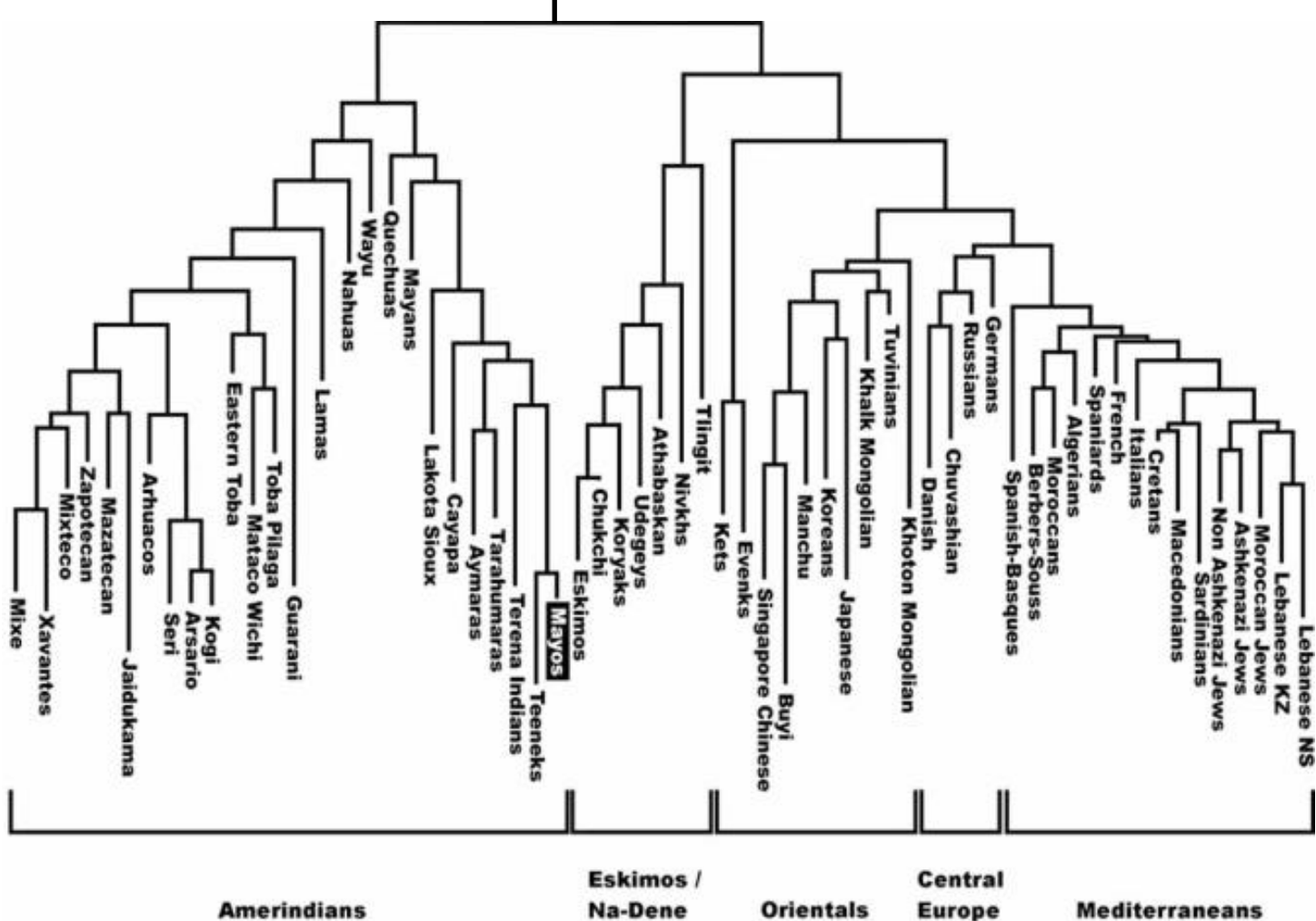
<https://www.youtube.com/watch?v=pIVk4NVIUh8>

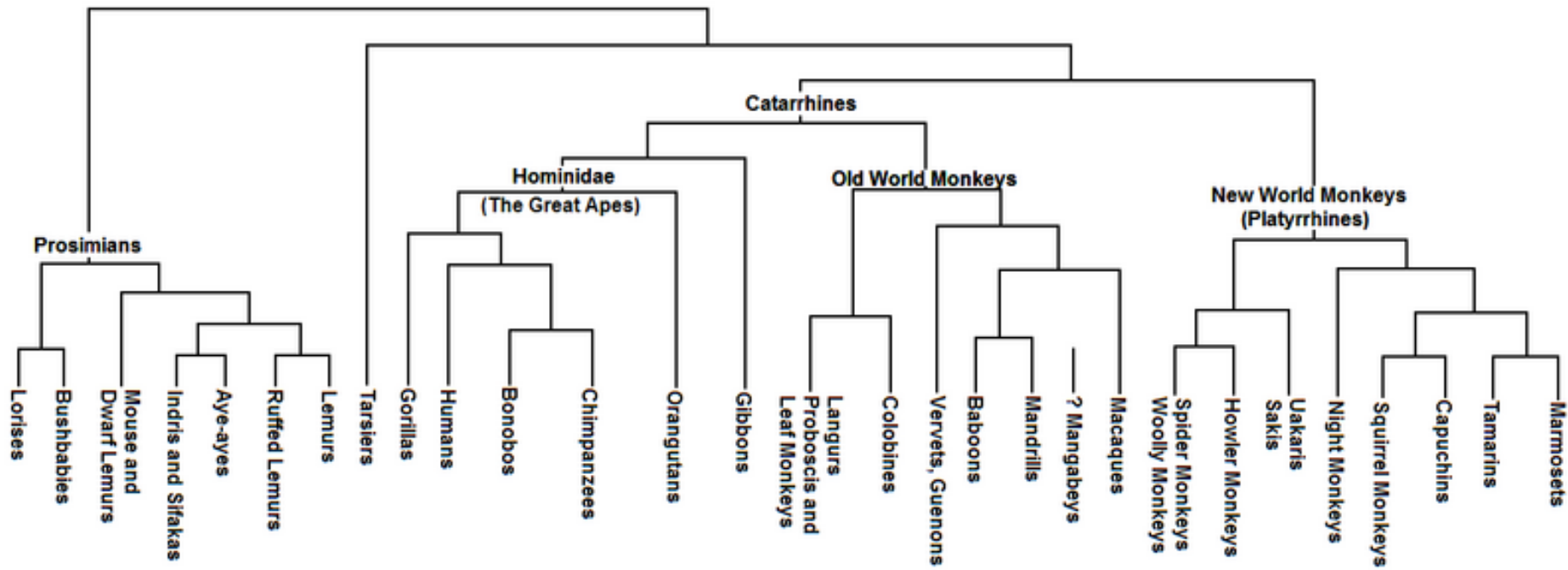


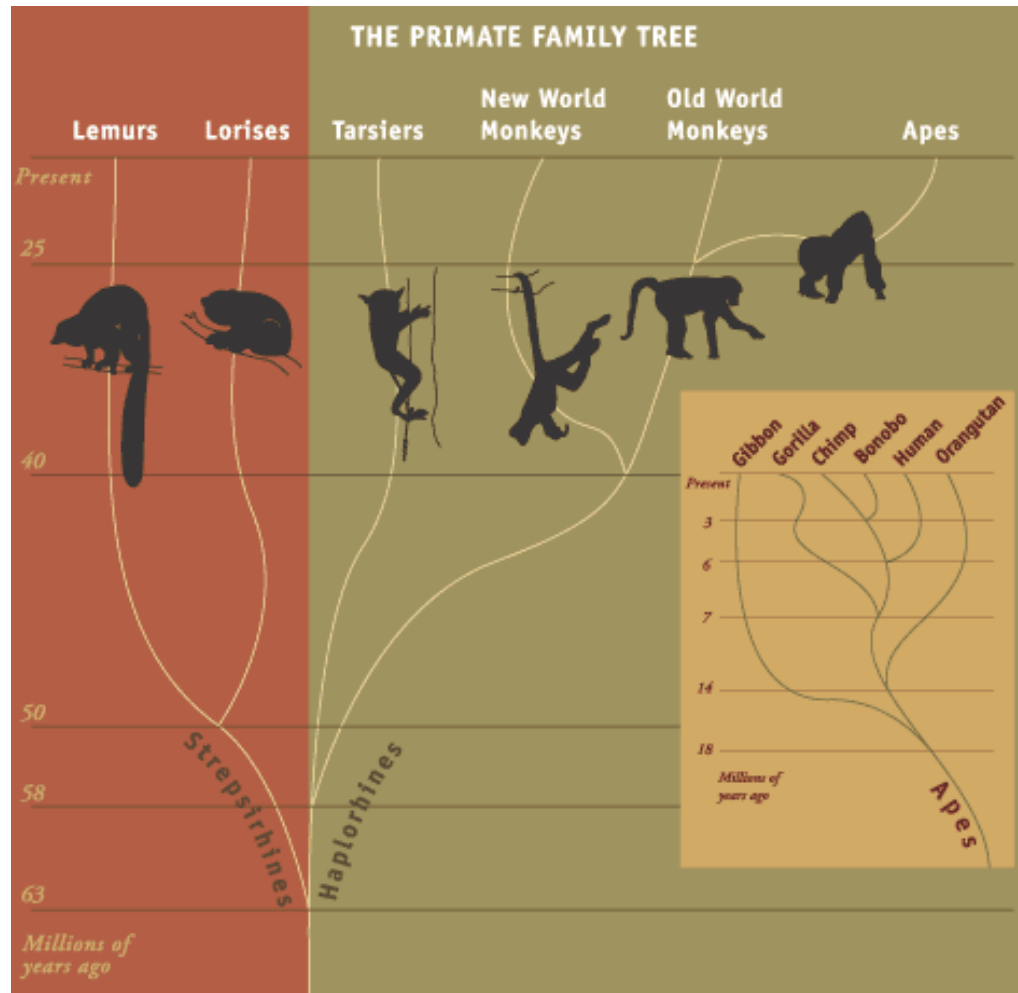


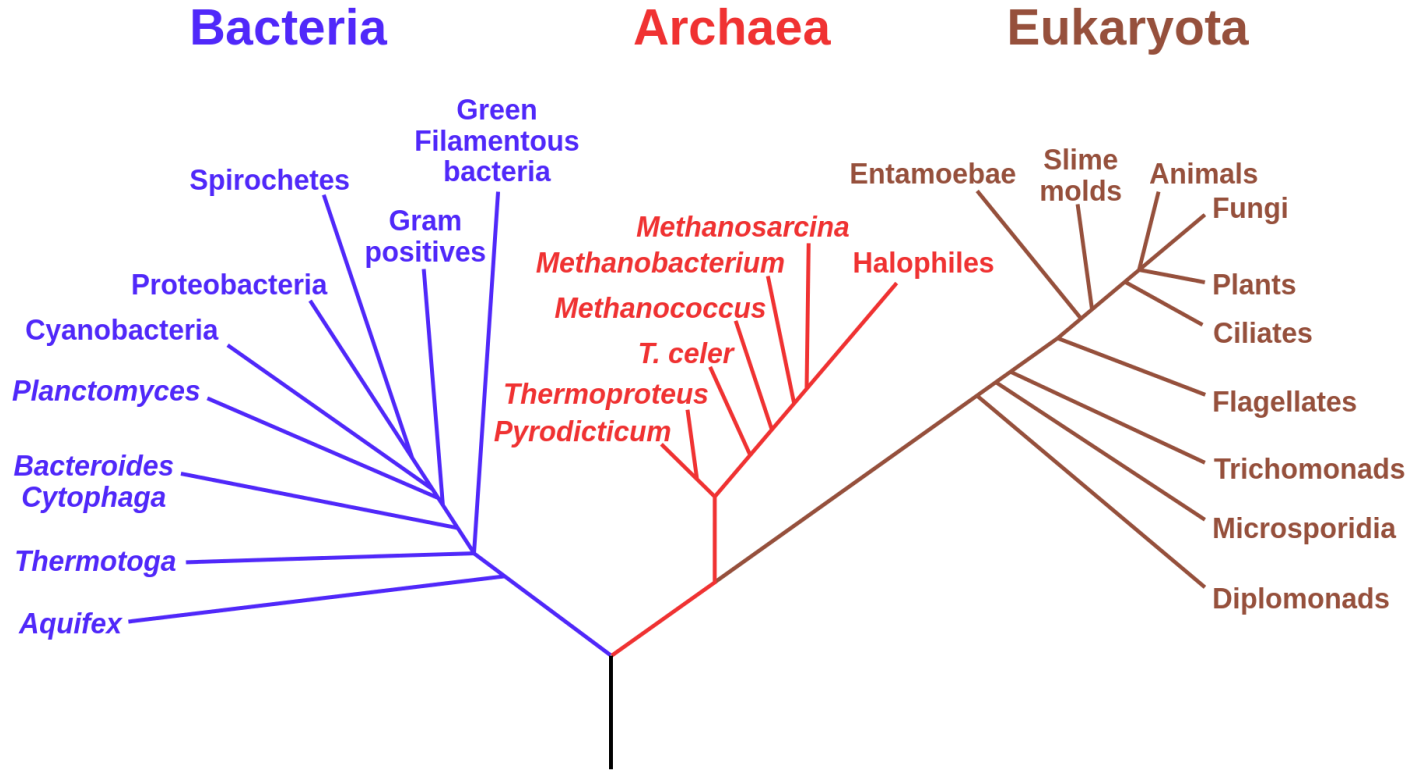




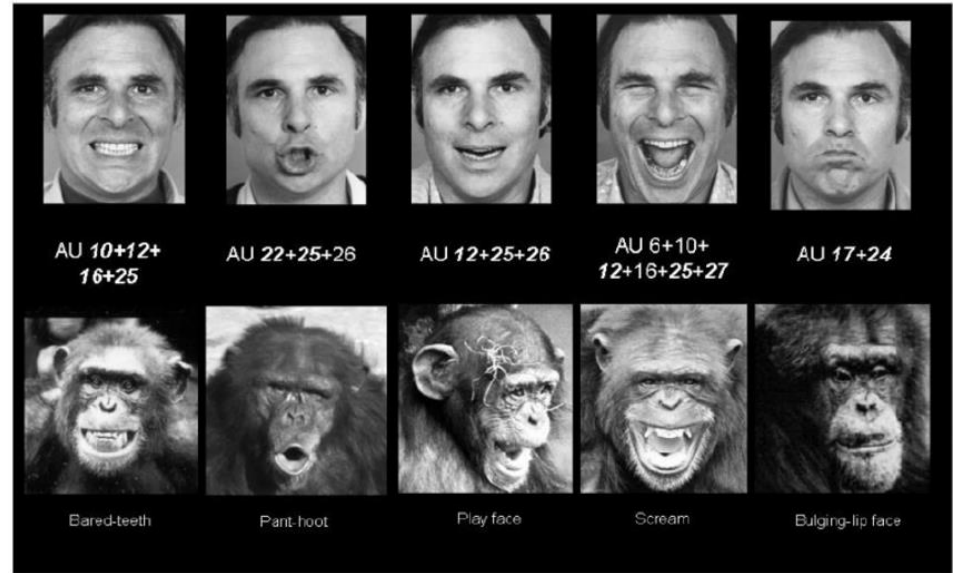
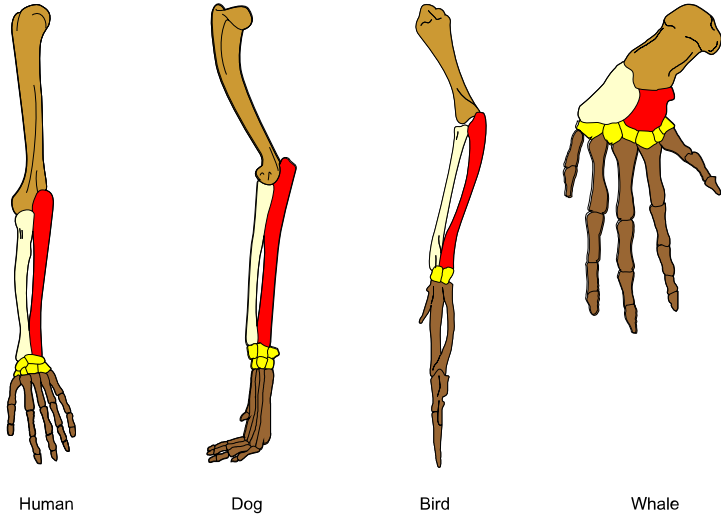






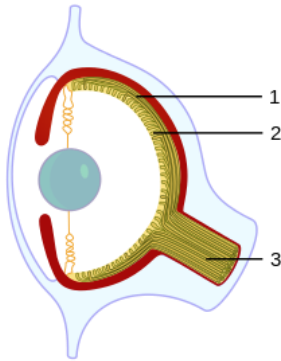
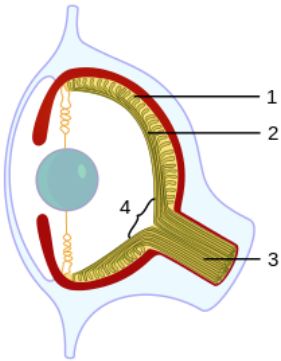


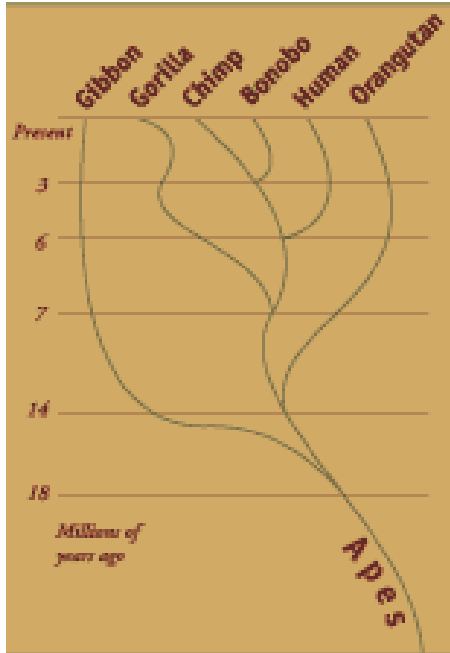
Homology



From Parr, L. A., & Waller, B. M. (2006). Understanding chimpanzee facial expression: insights into the evolution of communication. *SCAN*, 1, 221-228.

Convergent evolution





Humans and white-handed gibbons produce complex vocalisations featuring rule-governed combinations of a limited set of discrete sound types. Is this likely to be a homologous trait?

Go to wooclap.com

Use code INMRBS

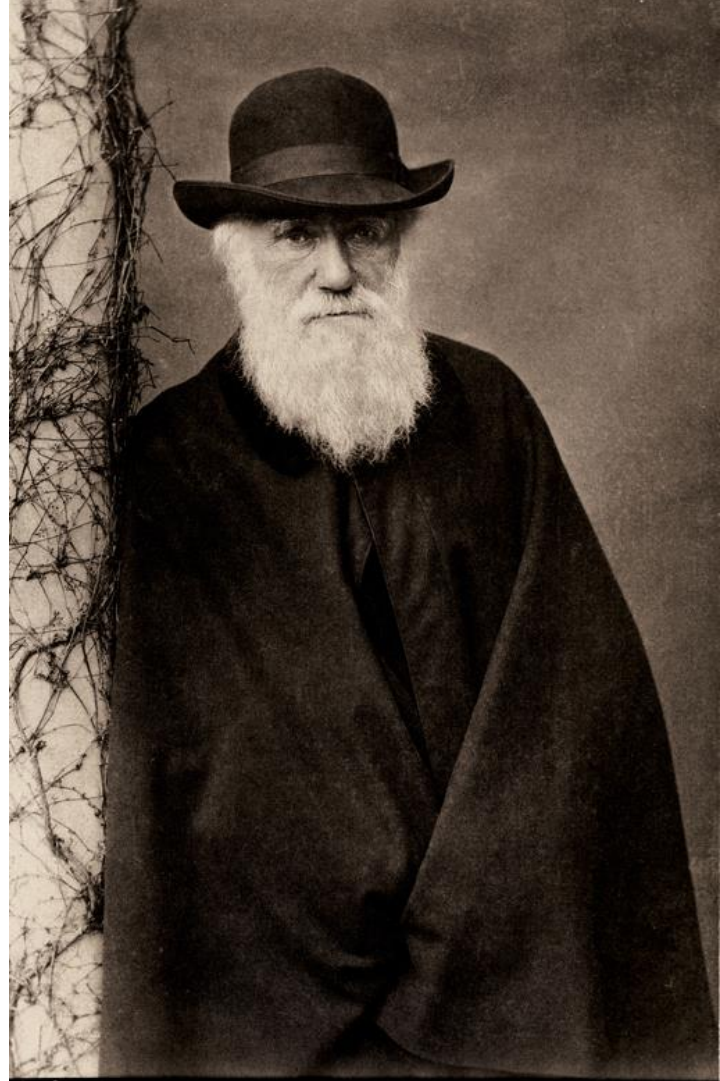
1: Yes

2: No

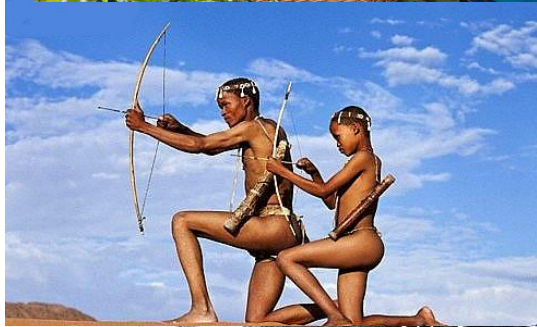
3: I need more information

Language as an adaptation?

- Language is just like any other **adaptive** feature of an organism's biology
- It's an **innate** feature of the human mind
- It **evolved by natural selection** under pressure for communication



A tool for the communication of knowledge and internal states



The core of the argument (from Pinker & Bloom, 1990)



“We will suggest that language shows signs of design for the communication of propositional structures over a serial channel.

The propositions in [human thought] are relational structures whose symbols pertain to people, objects, and events, the categories they belong to, their distribution in space and time, and their causal relations to one another ... The causal relations governing the behavior of other people are understood as involving their beliefs and desires, which can be reconsidered as relations between an individual and the proposition that represents the content of that belief or desire”

Pinker, S., & Bloom, P. (1990). Natural language and natural selection. *Behavioural and Brain Sciences*, 13, 707-784.
Lengthy quote is from p. 712-713

The core of the argument (from Pinker & Bloom, 1990)



“We would want to be able to

- refer to individuals and classes,*
- to distinguish among basic ontological categories (things, events, places, times, manners, and so on),*
- to talk about events and states, distinguishing the participants in the event or state according to role (agents, patients, goals),*
- to talk about the intentional states of ourselves and others*
- to express distinctions of truth value, modality (necessity, possibility, probability, factivity)*

Pinker, S., & Bloom, P. (1990). Natural language and natural selection. *Behavioural and Brain Sciences*, 13, 707-784.
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The core of the argument (from Pinker & Bloom, 1990)



- *to comment on the time of an event or state, including both its distribution over time (continuous, iterative, punctate) and its overall time of occurrence*
- *to encode an unlimited number of predicates, arguments, and propositions*
- *to be able to use the same propositional content within different speech acts; for instance, as a question, a statement, or a command.*
- *to focus or to put into the background different parts of a proposition, so as to tie the speech act into its context of previously conveyed information and patterns of knowledge of the listener. ”*

The core of the argument (from Pinker & Bloom, 1990)



“All we have argued is that human language, like other specialized biological systems, evolved by natural selection. Our conclusion is based on two facts that we would think would be entirely uncontroversial: Language shows signs of complex design for the communication of propositional structures, and the only explanation for the origin of organs with complex design is the process of natural selection.” (p. 726)

Are you persuaded? wooclap.com, code INMRBS

1: Yes

2: No

3: I need more information

Pinker, S., & Bloom, P. (1990). Natural language and natural selection. *Behavioural and Brain Sciences*, 13, 707-784.

Summary

- Adaptation & natural selection
 - A consequence of heritable variation in fitness
 - Similarity due to common ancestry (homology)
 - Convergent evolution and analogy
- Language as a biological adaptation

Next up

- Workshops
 - Attendance recorded and compulsory
 - Know where you are going
 - Take a look at the exercise in advance (less intensive prep this week than usual)
 - **Bring a laptop!**
- Next week's lecture: intention and structure in animal communication